



Motivation

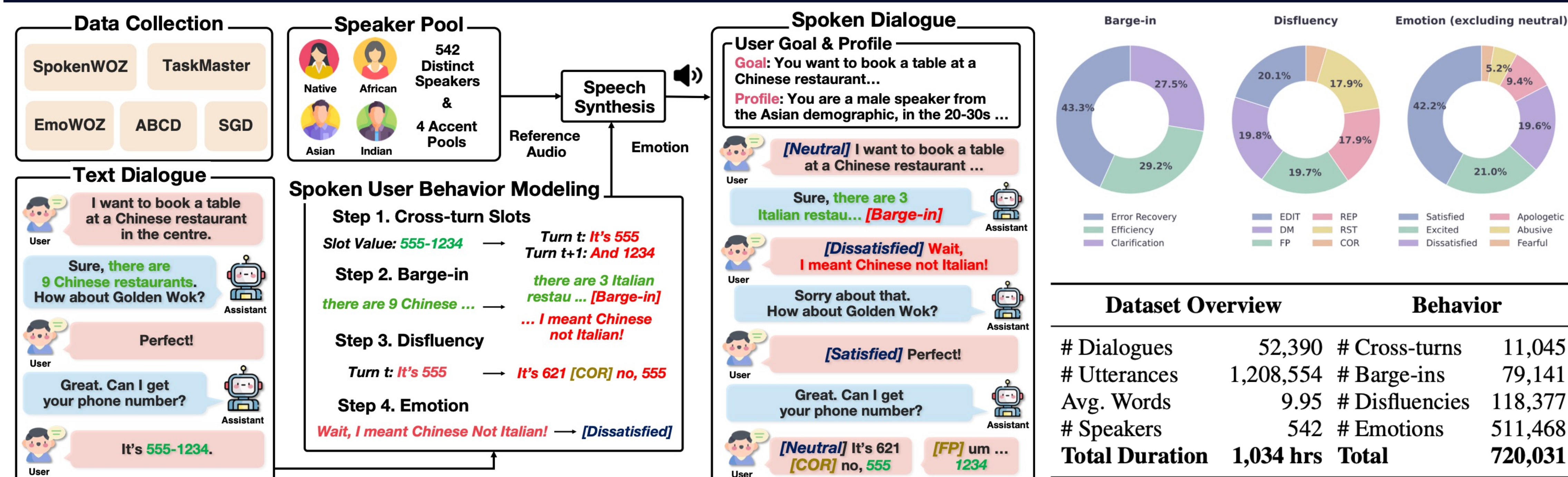
- Real users' speech is full of **spontaneous behaviors**, which are known to degrade agents' performance.
- User simulation is the scalable way to train and evaluate them, but existing simulators are **text-only**, and no spoken dialogue pipeline exists to build one.

💡 **SpokenTOD**: We augment text corpora with spoken behaviors and synthesize them into a large-scale spoken dialogue dataset.

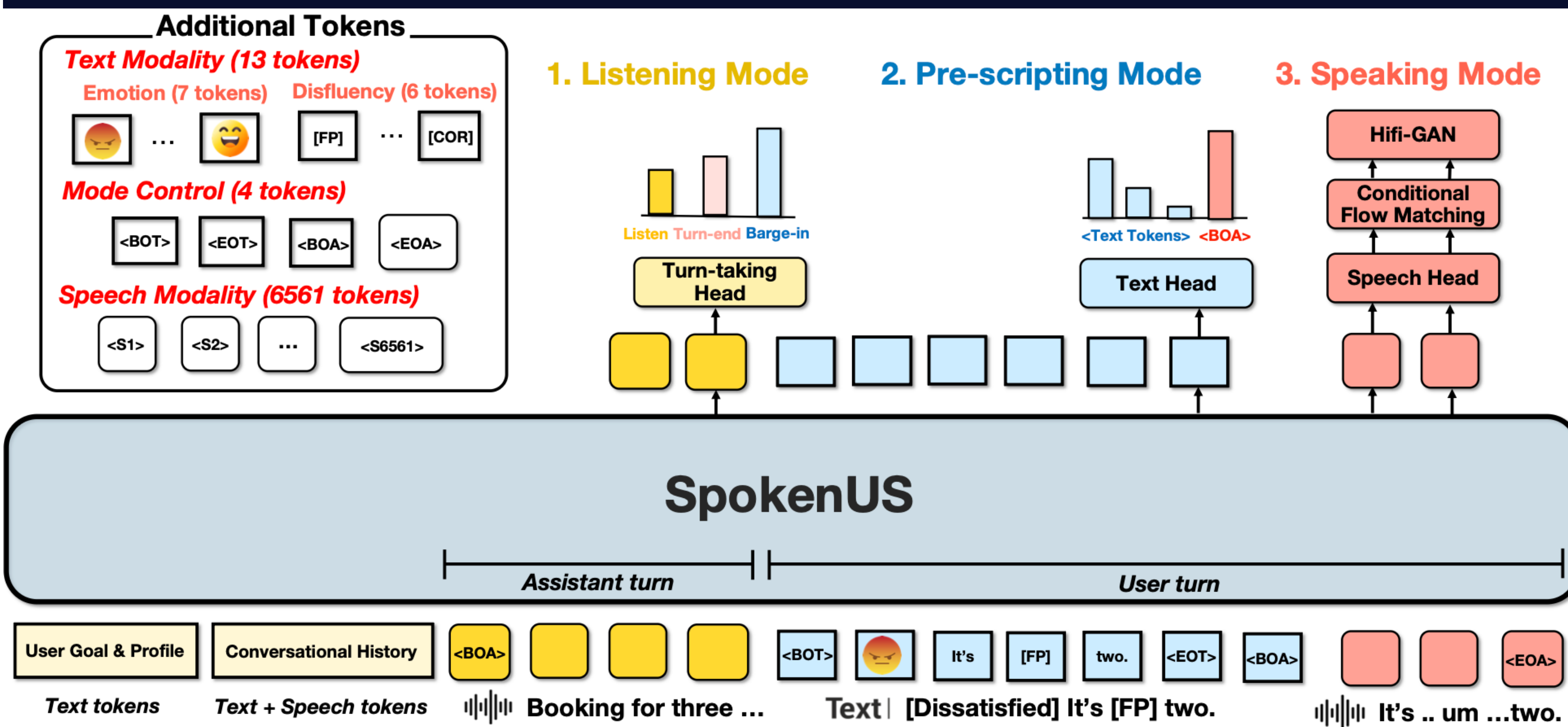
💡 **SpokenUS**: On top of it, we build a TOD-grounded spoken user simulator with proactive barge-in.

Category	Type	Definition
Cross-Turn Slots Phone numbers, email addresses, reservation IDs		
Barge-in	Error Recovery	Syst: "For two peo..." User: "No, three."
	Clarification	Syst: "It's nearby..." User: "Which address?"
	Efficiency	Syst: "Successfully..." User: "Thanks!"
Disfluency	Filled Pause	[FP] uh, we should go there.
	Discourse Marker	[DM] you know, the cheap one.
	Edit	reserve three, [EDIT] I mean, four people.
	Restart	book a ta- [RST] I'd like a table for two.
	Repetition	I want, [REP] I want a single room.
	Correction	left, [COR] no, right after the bank.
Emotion	Neutral, Fearful, Dissatisfied, Apologetic, Abusive, etc.	

SpokenTOD



SpokenUS



Training Setup

- Base Model**: Qwen2.5-3B
- Tokenizer**: FSQ from CosyVoice3
- Vocoder**: CFM + HiFi-GAN
- Objective**:

$$\mathcal{L} = \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{speech}} + \mathcal{L}_{\text{turn-taking}}$$

Experimental Setup

- Baselines**: Human recordings and 5 end-to-end omni models
- Metrics**: Goal Coverage (GA, SMR), Human MOS (Naturalness, Conversational Flow, Spoken User Behavior)
- Agent**: GPT-4.1-mini

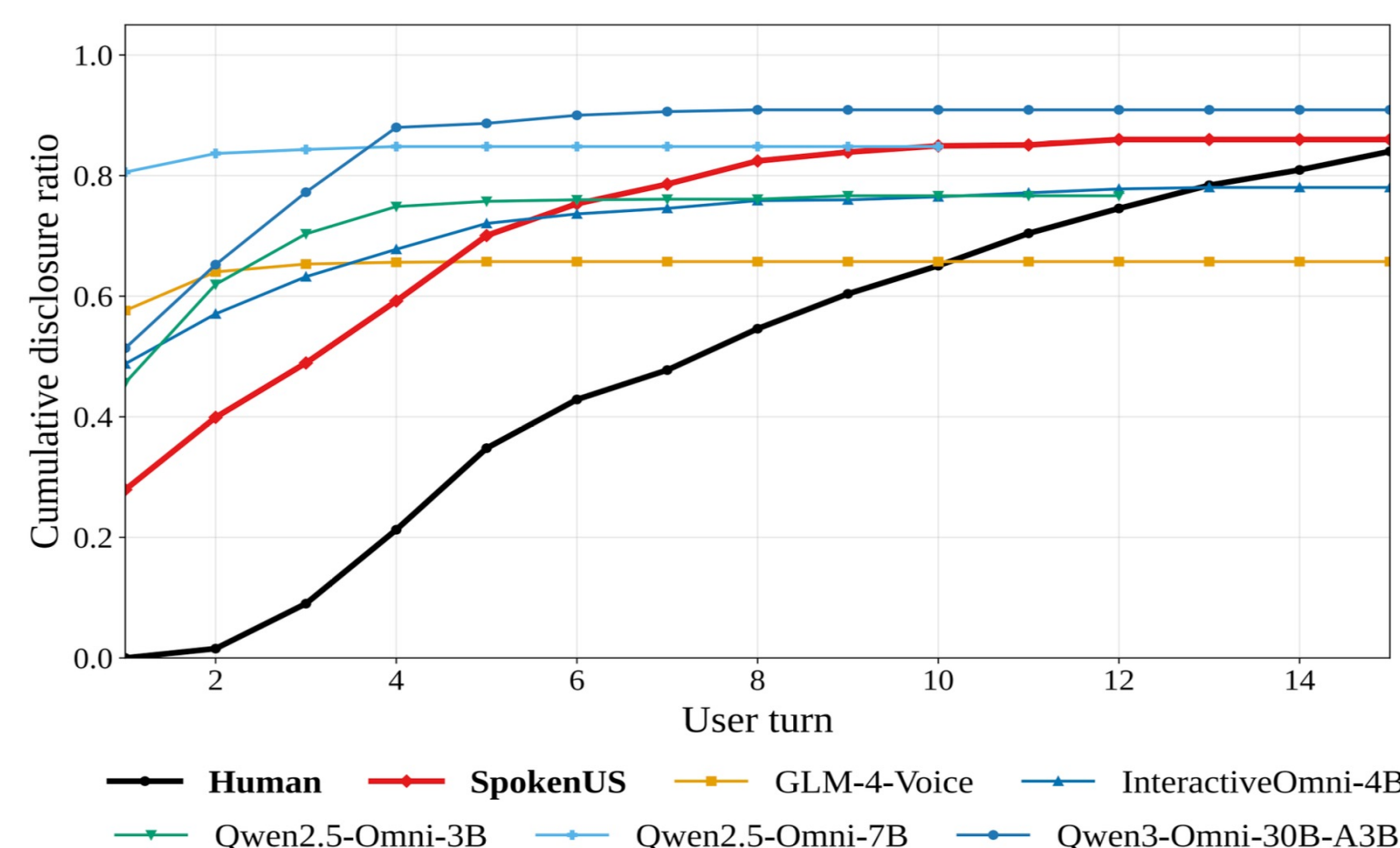
Experiments

SpokenUS vs. Baselines

- Is SpokenUS a more realistic user than baselines?

Model	Goal Cov.		Human MOS			
	GA	SMR	N	C	S	Avg.
Human	1.00	1.00	3.58	3.64	3.80	3.67
Larger models						
Qwen2.5-Omni-7B	0.80	0.97	2.86	3.29	2.94	3.03
GLM-4-Voice-9B	0.33	0.79	2.51	2.06	2.38	2.32
Qwen3-Omni-30B-A3B	0.93	0.99	3.03	3.61	2.91	3.18
Comparable-sized models						
Qwen2.5-Omni-3B	0.36	0.93	1.97	2.77	2.28	2.34
InteractiveOmni-4B	0.59	0.89	2.83	2.44	2.62	2.63
SpokenUS (3B)	0.82	0.99	3.92	4.14	4.11	4.06

- Does it disclose slots (the user goal's value) like humans?



- Does it challenge agents? Does it generalize across agents?

Model	Text	Cascaded	Assistant Model	Text	Cascaded
Qwen2.5-Omni-3B	86.8	83.1 ↓3.7	Qwen2.5-7B-Instruct	73.5	57.2 ↓16.3
SpokenUS	73.8	49.5 ↓24.3	Llama-3.1-8B-Instruct	65.8	45.7 ↓20.1